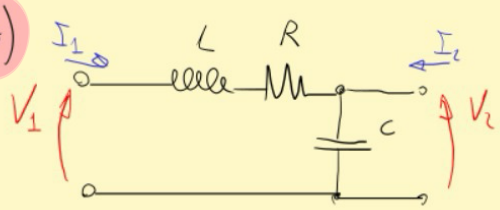
a)



impedancias: $Z_R = R$; $Z_L = sL$; $Z_C = \frac{1}{sC}$

→ L y R estarán en serie: $Z_1 = Z_L + Z_R = sL + R$

→ como la tensión de salida se mide sobre el capacitor $Z_2 = Z_C = \frac{1}{sC}$

→ usando el divisor de impedancias:

item 1

$$H(s) = \frac{Z_2}{Z_1 + Z_2} = \frac{1/sC}{sL + R + \frac{1}{sC}} = \frac{sC}{sC} \cdot \frac{1/sC}{sL + R + \frac{1}{sC}} = \frac{1}{s^2CL + sCR + 1} = \frac{1}{LC\left(s^2 + \frac{sR}{L} + \frac{1}{LC}\right)} = \frac{1}{LC\left(s^2 + \frac{sR}{L} + \frac{1}{LC}\right)} \cdot \frac{LC}{LC} = \frac{1/LC}{s^2 + s\frac{R}{L} + \frac{1}{LC}} \quad (*)$$

comparo con fórmula genral: $\frac{1/LC}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2}$, $\omega_0^2 = \frac{1}{LC}$, $\frac{\omega_0}{Q} = \frac{R}{L}$

entonces $\omega_0^2 = \frac{1}{LC}$ y $\frac{\omega_0}{Q} = \frac{R}{L}$

como $\omega_0 = 1 \Rightarrow 1^2 = \frac{1}{LC} \Rightarrow \boxed{LC = 1}$ (*)

como $\omega_0 = 1$, $R = 1$ y $Q = \frac{\sqrt{2}}{2} \Rightarrow \frac{2}{\sqrt{2}} = \frac{1}{L} \Rightarrow \boxed{L = \frac{\sqrt{2}}{2}}$ reemplazando en (*) $C \cdot \frac{\sqrt{2}}{2} = 1 \Rightarrow \boxed{C = \frac{2}{\sqrt{2}} = \sqrt{2}}$

volvemos a (*) y reemplazamos

$$H(s) = \frac{1/LC}{s^2 + s\frac{R}{L} + \frac{1}{LC}} = \frac{1}{s^2 + \sqrt{2}s + 1}$$

item 2 sabemos que $s = j\omega$ entonces

$$H(j\omega) = \frac{1}{(j\omega)^2 + \sqrt{2}j\omega + 1} = \frac{1}{-\omega^2 + \sqrt{2}\omega j + 1} = \frac{1}{\underbrace{1 - \omega^2}_{\text{real}} + \underbrace{\sqrt{2}\omega j}_{\text{imaginaria}}}$$

Prop: $\arg\left(\frac{N}{D}\right) = \arg(N) - \arg(D)$

$Tg(\theta) = \frac{b}{a} \rightarrow \theta = \arctg\left(\frac{b}{a}\right)$

$G_{dB}(\omega) = 20 \log_{10} |H(j\omega)|$

Calculamos el modulo: $|H(j\omega)| = \frac{1}{\sqrt{(1-\omega^2)^2 + (\sqrt{2}\omega)^2}} = \frac{1}{\sqrt{\omega^4 + 1}}$

$$|H(j.1)| = \frac{1}{\sqrt{1^4 + 1}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \Rightarrow \boxed{|H(j)| = \frac{\sqrt{2}}{2}}$$

Calculamos la fase: $\varphi(\omega) = \arg H(j\omega)$

$\varphi(\omega) = \arg(1) - \arg(1 - \omega^2 + \sqrt{2}\omega j)$

$\varphi(1) = -\arg(1 - 1 + \sqrt{2}j)$

$\boxed{\varphi(1) = -90°}$

Pasamos a dB: $G_{dB}(1) = 20 \log_{10}(\sqrt{2}/2) = -3.01\,dB$

Para graficar: modulo:

$\omega << 1$ $G_{dB}(\omega) = 20 \log_{10}(1) \approx 0\,dB$

$\omega >> 1$ $G_{dB}(\omega) = 20 \log_{10}\left(\frac{1}{\omega^2}\right) \approx 20(-2)\log_{10}(\omega) \approx -40 \log_{10}(\omega)$ → pendiente asintotica de -40 dB/decada.

fase: Partimos de $H(j\omega) = \frac{1}{(1-\omega^2) + j\sqrt{2}\omega}$

$\omega << 1 \Rightarrow H(j\omega) \approx 1$ es un numero real positivo su argumento es 0. $\varphi(\omega) = 0°$

$\omega >> 1 \Rightarrow H(j\omega) \approx \frac{1}{-\omega^2}$ es un numero real negativo entonces su argumento es 180°. $\varphi(\omega) = -180°$

Fase asintotica: $\varphi(\omega) = -\arctg\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$

$\lim_{\omega \to 0} -\arctg\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = -\arctg(0) = 0$

$\lim_{\omega \to 1} -\arctg\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = -\arctg\left(\frac{\sqrt{2}}{0}\right) = -90°$

$\lim_{\omega \to \infty} -\arctg\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = -\arctg(-0) = -180°$

GRAFICOS

ítem 3 Para saber que tipo de filtro es, analizamos cuando la frecuencia es muy baja y cuando es muy alta.

$\lim_{\omega \to 0} |H(j\omega)| = 1$ $\lim_{\omega \to \infty} |H(j\omega)| = 0$, Entonces es un FILTRO PASA-BAJOS DE SEGUNDO ORDEN.

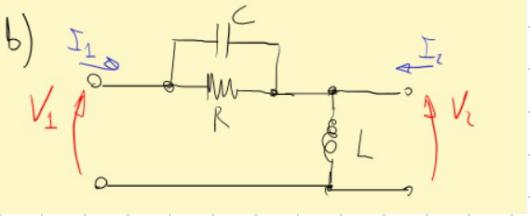
El ancho de banda comienza en la frecuencia 0 y termina en la frecuencia de corte, entonces $BW = \omega_c - 0 = \omega_c$

Calculo frecuencia de corte:

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}} \quad \text{igualamos} \quad \frac{1}{\sqrt{1+\omega_c^4}} = \frac{1}{\sqrt{2}} \Rightarrow 1+\omega_c^4 = 2$$

$$\boxed{\omega_c = 1 \text{ rad/s}}$$

b)



C y R en paralelo $Z_1 = \dfrac{R \cdot \frac{1}{sC}}{\frac{1}{sC} + R} = \dfrac{R}{1 + sRC}$

$Z_2 = Z_L = sL$

item 1

$$H(s) = \frac{Z_2}{Z_1 + Z_2} = \frac{sL}{\frac{R}{1+sRC} + sL} = \frac{sL}{\frac{R + sL(1+sRC)}{1+sRC}} = \frac{sL + s^2LRC}{R + sL + s^2LRC} = \frac{LRC}{LRC} \cdot \frac{s^2 + s.\frac{1}{RC}}{s^2 + s.\frac{1}{RC} + \frac{1}{LC}} \quad (*)$$

($\frac{w_0}{Q}$ sobre $s.\frac{1}{RC}$; w_0^2 sobre $\frac{1}{LC}$)

$w_0^2 = \dfrac{1}{LC} \Rightarrow LC = 1$

$\dfrac{w_0}{Q} = \dfrac{1}{RC} \Rightarrow \dfrac{2}{\sqrt{2}} = \dfrac{1}{C} \Rightarrow \boxed{C = \dfrac{\sqrt{2}}{2}}$ entonces $\boxed{L = 2}$

volvemos a $(*)$ y reemplazamos

$$H(s) = \frac{s^2 + s.\frac{1}{RC}}{s^2 + s.\frac{1}{RC} + \frac{1}{LC}} = \frac{s^2 + \sqrt{2}s}{s^2 + \sqrt{2}s + 1}$$

item 2

sabemos que $s = jw$ entonces

$$H(jw) = \frac{(jw)^2 + \sqrt{2}(jw)}{(jw)^2 + \sqrt{2}(jw) + 1} = \frac{-w^2 + j\sqrt{2}w}{-w^2 + j\sqrt{2}w + 1} = \frac{-w^2 + j\sqrt{2}w}{1 - w^2 + j\sqrt{2}w}$$

Calculamos el modulo: $|H(jw)| = \dfrac{\sqrt{(-w^2)^2 + (\sqrt{2}w)^2}}{\sqrt{(1-w^2)^2 + (\sqrt{2}w)^2}} = \dfrac{\sqrt{w^4 + 2w^2}}{\sqrt{1 + w^4}} = \dfrac{w\sqrt{w^2 + 2}}{\sqrt{1 + w^4}}$ con $w \geq 0$

$|H(j)| = \dfrac{\sqrt{1^2 + 2}}{\sqrt{1+1}} = \sqrt{\dfrac{3}{2}}$ $\quad G_{dB}(1) = 20.\log_{10}\left(\sqrt{\frac{3}{2}}\right) \approx 1{,}76$ dB.

Para $w \ll 1 \Rightarrow \lim_{w \to 0} |H(jw)| = 0$

$G_{dB}(w) = 20.\log_{10}(\sqrt{2}) + 20\log_{10}(w) \approx 3{,}01 + 20\log_{10}(w)$ → Pendiente asintota de 20dB/decada

Para $w \gg 1 \Rightarrow \lim_{w \to \infty} |H(jw)| = 1$ $\quad G_{dB}(w) \approx 0$ dB

Calculamos la fase:

$\varphi(w) = \arg H(jw)$

$\varphi(w) = \arg(-w^2 + j\sqrt{2}w) - \arg(1 - w^2 + j\sqrt{2}w)$

Para $w = 1$ $\quad \varphi(1) = \arg(-1^2 + j\sqrt{2}.1) - \arg(1 - 1^2 + j\sqrt{2}.1)$

$\varphi(1) = \underbrace{\arg(-1 + \sqrt{2}j)}_{\text{II cuadrante}} - \underbrace{\arg(\sqrt{2}j)}_{\text{eje imaginario}}$

$\varphi(1) = \pi - \operatorname{arctg}(\sqrt{2}) - \frac{\pi}{2} = 90° - \operatorname{arctg}(\sqrt{2}) \approx 35{,}26$

Propiedad:

$\arg\left(\frac{A.B}{C}\right) = \arg(A) + \arg(B) - \arg(C)$

Para $w \ll 1$ $\quad \varphi(w) = \arg(-w^2 + j\sqrt{2}w) - \arg(1 - w^2 + j\sqrt{2}w)$

$\varphi(w) = \pi - \operatorname{arctg}\left(\frac{\sqrt{2}w}{w^2}\right) - \operatorname{arctg}\left(\frac{\sqrt{2}w}{1-w^2}\right)$

$\lim_{w \to 0}$ $\varphi(w) = \pi - \operatorname{arctg}\left(\frac{\sqrt{2}}{w}\right) - \operatorname{arctg}\left(\frac{\sqrt{2}w}{1-w^2}\right)$ ($\frac{\sqrt{2}}{w} \to \infty$)

$\varphi(w) = \pi - \frac{\pi}{2} - 0 = \frac{\pi}{2} = 90°$

Para $w \gg 1$ $\quad \varphi(w) = \arg(-w^2 + j\sqrt{2}w) - \arg(1 - w^2 + j\sqrt{2}w)$

$\varphi(w) = \pi - \operatorname{arctg}\left(\frac{\sqrt{2}w}{w^2}\right) - \pi + \operatorname{arctg}\left(\frac{\sqrt{2}w}{1-w^2}\right)$

$\lim_{w \to 0}$ $\varphi(w) = \pi - \operatorname{arctg}\left(\frac{\sqrt{2}}{w}\right) - \pi + \operatorname{arctg}\left(\frac{\sqrt{2}w}{1-w^2}\right)$ ($\frac{\sqrt{2}}{w} \to 0$, $\frac{\sqrt{2}w}{1-w^2} \to 0$)

$\varphi(w) = \pi - \pi = 0$

Buscar la fase asintotica: $\varphi(\omega) = \text{arctg}\left(\frac{\sqrt{2}}{-\omega}\right) - \text{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$

$\lim_{\omega \to 0} -\text{arctg}\left(\frac{\sqrt{2}}{-\omega}\right) - \text{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = 90°$

$\lim_{\omega \to 1} -\text{arctg}\left(\frac{\sqrt{2}}{-\omega}\right) - \text{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = \pi - \text{arctg}(\sqrt{2}) - \frac{\pi}{2} = 90° - \text{arctg}(\sqrt{2})$

$\lim_{\omega \to \infty} -\text{arctg}\left(\frac{\sqrt{2}}{-\omega}\right) - \text{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = 0°$

GRAFICOS

item 3 veo que pasa a muy bajas y muy altas frecuencias.

$\lim_{\omega \to 0} |H(j\omega)| = 0$ $\lim_{\omega \to \infty} |H(j\omega)| = 1$, entonces es un FILTRO PASA-ALTOS DE SEGUNDO ORDEN.

El ancho de banda comienza en la frecuencia de corte y se extiende hacia las frecuencias altas, entonces

Calculo frecuencia de corte:

$|H(j\omega_c)| = \frac{1}{\sqrt{2}}$ igualamos $\frac{1}{\sqrt{2}} = \frac{\omega_c \sqrt{\omega_c^2 + 2}}{\sqrt{1+\omega_c^4}}$

$\frac{1}{2} = \frac{\omega_c^2(\omega_c^2+2)}{1+\omega_c^4} = \frac{\omega_c^4 + 2\omega_c^2}{1+\omega_c^4}$

$1 + \omega_c^4 = 2\omega_c^4 + 4\omega_c^2$

$1 = \omega_c^4 + 4\omega_c^2$ ⟶ SUSTITUCION $u = \omega_c^2$

$0 = u^2 + u + 1$

$u \longrightarrow \omega_c^2 = -2 - \sqrt{5}$ abs!

$\omega_c^2 = -2 + \sqrt{5}$

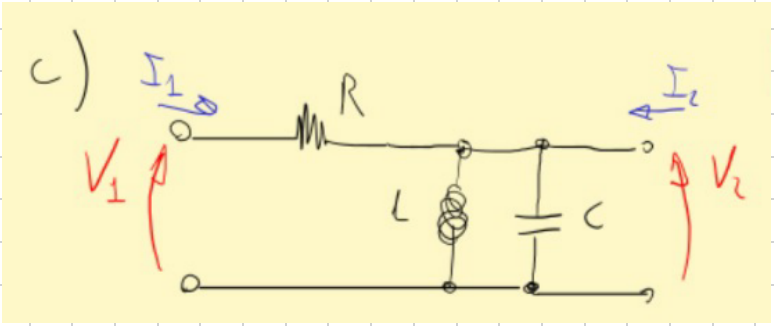
$\omega_c \longrightarrow 0{,}48$

$-0{,}48$ abs!

$BW = \boxed{\omega_{superior}} - \omega_c$ $\boxed{\omega_{superior}} = \infty$ ya que es un modelo ideal y no establece un limite superior para la banda pasante.

$BW = \infty - 0{,}48 = \infty$

c)



$Z_1 = R$

L y C estarán en paralelo.

$$Z_2 = \frac{SL \cdot \frac{1}{SC}}{SL + \frac{1}{SC}} = \frac{L/C}{SL + 1/SC} \cdot \frac{SC}{SC} = \frac{SL}{1+s^2LC}$$

ítem 1

$$H(s) = \frac{\frac{SL}{1+s^2LC}}{R + \frac{SL}{1+s^2LC}} \cdot \frac{1+s^2LC}{1+s^2LC} = \frac{SL}{R+s^2LCR+SL} \cdot \frac{\frac{1}{LCR}}{\frac{1}{LCR}} = \frac{s \cdot \frac{1}{CR}}{s^2 + s\frac{1}{CR} + \frac{1}{LC}}$$

($\frac{1}{CR} = \frac{\omega_0}{Q}$, $\frac{1}{LC} = \omega_0^2$)

entonces si $\frac{1}{CR} = \frac{\omega_0}{Q}$ y $\frac{1}{LC} = \omega_0^2$

con $\omega = 1$ $\quad 1^2 = \frac{1}{LC} \Rightarrow LC = 1$

$$\frac{1}{C} = \frac{2}{\sqrt{2}} \Rightarrow \boxed{C = \frac{\sqrt{2}}{2}} \quad \Rightarrow \boxed{L = \sqrt{2}}$$

reemplazamos con los valores obtenidos $H(s) = \frac{\sqrt{2}s}{s^2+\sqrt{2}s+1}$

ítem 2 Sabemos que $s = j\omega$ entonces $H(j\omega) = \frac{\sqrt{2}j\omega}{-\omega^2+\sqrt{2}j\omega+1} = \frac{\sqrt{2}\omega j}{1-\omega^2+\sqrt{2}\omega j}$

Calculamos el modulo $|H(j\omega)| = \frac{\sqrt{(\sqrt{2}\omega)^2}}{\sqrt{(1-\omega^2)^2+(\sqrt{2}\omega)^2}} = \frac{\sqrt{2}\omega}{\sqrt{1+\omega^4-2\omega^2+2\omega^2}} = \frac{\sqrt{2}\omega}{\sqrt{1+\omega^4}}$

Para $\omega = 1$ $\quad |H(j)| = \frac{\sqrt{2}}{\sqrt{2}} = 1$ $\quad G_{dB}(\omega) = 20\log_{10}(1) = 0\,dB$

Para $\omega \ll 1$ $\quad |H(j\omega)| = \sqrt{2}\omega$ $\quad G_{dB} = 20\log_{10}(\sqrt{2}) + 20\log_{10}(\omega) = 3.01 + 20\log_{10}(\omega)$ → Pendiente asintótica de 20dB/decada

Para $\omega \gg 1$ $\quad |H(j\omega)| = \frac{\sqrt{2}\omega}{\sqrt{\omega^4}} = \frac{\sqrt{2}}{\omega}$ $\quad G_{dB} = 20\log_{10}(\sqrt{2}) - 20\log_{10}(\omega) = 3{,}01 - 20\log_{10}(\omega)$ → Pendiente asintótica de -20dB/decada

Calculamos la fase:

$$\varphi(\omega) = \arg(H(j\omega))$$
$$\varphi(\omega) = \arg(\sqrt{2}\omega j) - \arg(1-\omega^2+\sqrt{2}\omega j)$$
$$\varphi(\omega) = 90° - \arg(1-\omega^2+\sqrt{2}\omega j)$$

Para $\omega = 1$ $\quad \varphi(1) = 90° - \operatorname{arctg}\left(\frac{\sqrt{2}}{0}\right) = 0$

Para $\omega \ll 1$ $\quad \varphi(\omega) = 90° - \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$ $\quad \lim_{\omega \to 0} \varphi(\omega) = 90° - \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = 90°$ ($1-\omega^2 \to 0$)

Para $\omega \gg 1$ $\quad \varphi(\omega) = 90° - \pi + \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$ $\quad \lim_{\omega \to \infty} \varphi(\omega) = 90° - \pi + \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{\omega^2}\right) = -90°$ ($\frac{\sqrt{2}\omega}{\omega^2} \to 0$)

Buscar la fase asintótica: $\varphi(\omega) = 90° - \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right)$

$$\lim_{\omega \to 0} \; 90° - \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = 90°$$

$$\lim_{\omega \to 1} \; 90° - \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = 0$$

$$\lim_{\omega \to \infty} \; 90° - \operatorname{arctg}\left(\frac{\sqrt{2}\omega}{1-\omega^2}\right) = -90°$$

GRAFICOS

ítem 3 veo que pasa a muy bajas y muy altas frecuencias.

$\lim_{w \to 0} |H(jw)| = 0$ $\lim_{w \to \infty} |H(jw)| = 0$, Pero en $w=1$ $H(j) = 1$ entonces es un FILTRO PASA-BANDA DE SEGUNDO ORDEN.

El ancho de banda de un filtro pasa-banda tiene un limite inferior y superior. $BW = W_{c2} - W_{c1}$

$|H(jw_c)| = \frac{1}{\sqrt{2}}$ igualamos $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}\, w_c}{\sqrt{1+w_c^2}}$

$$\frac{1}{2} = \frac{2 w_c^2}{1 + w_c^4}$$

$$1 + w_c^4 = 4 w_c^2$$

$$w_c^4 - 4 w_c^2 + 1 = 0 \longrightarrow \text{sustitución } u = w_c^2$$

$$u^2 - 4u + 1 = 0$$

$u \longrightarrow 2 + \sqrt{3} = w_c^2 \longrightarrow 1{,}93 = w_{c2}$

$\qquad\qquad\qquad\quad \longrightarrow -1{,}93$ abs!

$u \longrightarrow 2 - \sqrt{3} = w_c^2 \longrightarrow 0{,}51 = w_{c1}$

$\qquad\qquad\qquad\quad \longrightarrow -0{,}51$ abs!

$BW = 1{,}93 - 0{,}51 = 1{,}42$